

APEX RETAIL AI

Store Intelligence & Customer Conversion Platform

Hackathon Submission Presentation

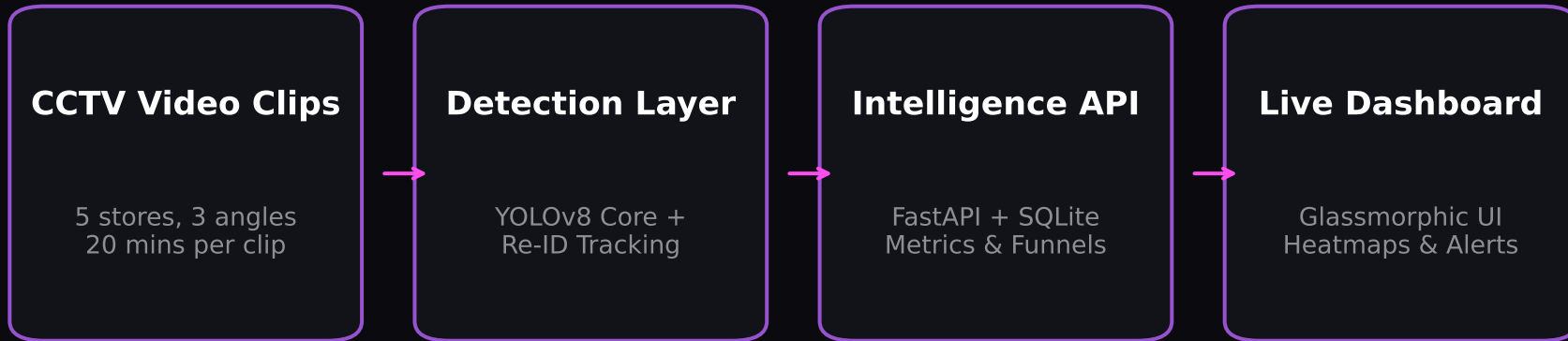
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THE OFFLINE RETAIL BLIND SPOT

Why Physical Retail Needs Online-Style Analytics:

- **No Path Visibility :** Physical stores have historically operated as a data blind spot. Customer journeys, zone dwells, and browsing paths remain completely untracked.
- **Checkout Drop-offs :** No systematic way to quantify how many shoppers enter the billing queue but abandon it due to long wait times.
- **Disconnect from Sales :** Total foot traffic is rarely linked back to POS transaction records, making it impossible to compute exact shopper conversion rates.
- **Inflexible Infrastructure :** Existing tracking solutions require heavy database setup and proprietary hardware, leading to complex and expensive deployments.

END-TO-END SYSTEM ARCHITECTURE



Key Technologies Used: Python 3.11 | Ultralytics YOLOv8 | FastAPI | SQLite3 | HTML5 & Vanilla CSS

KEY TECHNICAL SOLUTIONS

- ✓ **Strict Time Correlation**

Correlates a visitor's checkout events to a POS transaction strictly if a transaction occurred within a 5-minute window following their presence. Resolves transaction-buyer mapping without customer identity data.

- ✓ **Intelligent Staff Filter**

Excludes retail staff from customer conversion metrics using a color-histogram uniform classifier in the main floor tracking camera, tagging staff events with 'is_staff=true'.

- ✓ **Idempotency Safeguard**

Enforces event deduplication on API ingest using SQLite's native 'INSERT OR IGNORE' primary key index, ensuring safe, duplicate-free metrics updates on network retries.

- ✓ **Self-Contained Edge Build**

Designed using serverless SQLite database, eliminating manual setup and service dependency issues on edge devices, allowing one-click Docker deployments.

PRODUCTION READINESS & LIVE DASHBOARD

Live Glassmorphic Dashboard Production Readiness Metrics

- 100% viewport-contained single-page layout (no scrolling)
- Dynamic Real-Time zone heatmap overlays (frequency scaled by zone)
- Active Conversion Funnel tracking Entry -> Zone Visit -> Queue -> DB -> Purchase
- Live operational logs and active alerts for queues and high zones
- Zero Setup Containerization (runs via 'docker compose up')
- Structured 100% CDN logging tracing trace_id, store_id, latency, etc.
- Queue DB degradation mapping failures to HTTP 503 response
- High code coverage (>70% statement coverage in test suite)

Thank you! Live Demo at: purple-store-intelligence-xxm1.onrender.com